

COMPUTER SCIENCE & IT



Digital Logic



BOOLEAN THEOREMS AND GATES -1 Lecture : 01



By- Chandan Gupta Sir



Recap of Previous Lecture

NA.



Topics to be Covered

Theorems & some imp. basics.

gmp basics

• base = 10 $\longrightarrow$ (0-9)

$$\begin{matrix} 10^2 & 10^1 & 10^0 \\ (7 & 8 & 6) \end{matrix}_{10} = (7 \times 10^2 + 8 \times 10 + 6 \times 10^0)$$

$$(19)_{10}$$

$$(10)_{10} \longrightarrow \text{is number not a digit}$$

binary no. system:

$$\text{base} = (2) \longrightarrow (0-1)$$

$$\begin{matrix} 2^2 & 2^1 & 2^0 \\ (1 & 1 & 0) \end{matrix}_2 = (1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0)_{10} = (6)_{10}$$

$$(121)_2 \times$$

$$\begin{matrix} 2^3 & 2^2 & 2^1 & 2^0 \\ (1 & 1 & 0 & 1) \end{matrix}_2 = (13)_{10}$$

$$(101011)_2 = (32 + 8 + 2 + 1)_{10} = (43)_{10}$$

$$(1101010)_2 = (106)_{10}$$

$$4\text{-bits} \rightarrow 0000 \rightarrow (0)_{10}$$

⋮

$$(1111)_2 = (2^4 - 1)_{10} = (15)_{10}$$

$$(9999)_{10} = (10^4 - 1)_{10}$$

$$\begin{matrix} 8^3 & 8^2 & 8^1 & 8^0 \\ (7777)_8 = (8^4 - 1)_{10} \end{matrix}$$

$$[(x-1)(x-1)(x-1)(x-1)]_x = (x^4 - 1)_{10}$$

$$(24)_{10} = \begin{matrix} 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ (1 & 1 & 0 & 0 & 0) \end{matrix}_2$$

$$(43)_{10} = \begin{matrix} 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ (1 & 0 & 1 & 0 & 1 & 1) \end{matrix}_2$$

$$(54)_{10} = (110110)_2$$

$$(89)_{10} = (1011001)_2$$



2^1	2^0	
A	B	
0	0	$\longrightarrow (0)_{10}$
0	1	$\longrightarrow (1)_{10}$
1	0	$\longrightarrow (2)_{10}$
1	1	$\longrightarrow (3)_{10}$

$f(A, B)$

2^2	2^1	2^0	
A	B	C	
0	0	0	$\longrightarrow (0)_{10}$
0	0	1	$\longrightarrow (1)_{10}$
0	1	0	$\longrightarrow (2)_{10}$
0	1	1	$\longrightarrow (3)_{10}$
1	0	0	$\longrightarrow (4)_{10}$
1	0	1	$\longrightarrow (5)_{10}$
1	1	0	$\longrightarrow (6)_{10}$
1	1	1	$\longrightarrow (7)_{10}$

A	B	C	D	
0	0	0	0	$\longrightarrow (0)_{10}$
0	0	0	1	$\longrightarrow (1)_{10}$
0	0	1	0	$\longrightarrow (2)_{10}$
0	0	1	1	$\longrightarrow (3)_{10}$
0	1	0	0	$\longrightarrow (4)_{10}$
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	$\longrightarrow (15)_{10}$

n -variables $\longrightarrow N = 2^n$ comb.

$\longrightarrow 0 - (2^n - 1)_{10}$



[Boolean Theorems]



1. Related to 'OR' operation (+)

A	NOT	$\bar{A}$
0		1
1		0

• $A + 0 = A \rightarrow 0 + A = A$

$$\bar{A} + 0 = \bar{A}$$

$$AB + 0 = AB$$

$$A + B = B + A$$

• $A + A = A$

$$AB + AB = AB$$

$$A + A + A = A$$

$$\bar{A} + \bar{A} = \bar{A}$$

• $A + 1 = 1 = 1 + A$

$$\bar{A} + 1 = 1$$

$$AB + 1 = 1$$

$$1 + \underbrace{\bar{A}B + BC + AC}_{\text{gmp}} = 1$$

$$(1 + \text{anything}) = 1$$

• $A + \bar{A} = \bar{A} + A = 1$

• $AB + \overline{AB} = 1$

$$AB + \bar{A} \cdot \bar{B} \neq 1$$

• $0 + 0 = 0$

• $0 + 1 = 1$

• $1 + 0 = 1$

• $1 + 1 = 1$

logical
'OR'

$$\overline{A \cdot B} = (\bar{A} + \bar{B})$$

2. Related to 'AND' operation ($\cdot$)

$$A \cdot B = B \cdot A$$

- $A \cdot 0 = 0$

$$\overline{A} \cdot 0 = 0$$

$$AB \cdot 0 = 0$$

$$A \cdot B \cdot 0 \cdot C \cdot \overline{D} = 0$$

- $A \cdot \overline{A} = \overline{A} \cdot A = 0$

- $AB \cdot \overline{AB} = 0$

- $A \cdot 1 = 1 \cdot A = A$

$$\overline{A} \cdot 1 = \overline{A}$$

$$AB \cdot 1 = AB$$

$$A \cdot B \cdot C \cdot \overline{B} \cdot D \cdot E = 0$$


- $0 \cdot 0 = 0$

- $0 \cdot 1 = 0$

- $1 \cdot 0 = 0$

- $1 \cdot 1 = 1$

- $A \cdot A = A$

$$A \cdot A \cdot A = A$$

$$\overline{A} \cdot \overline{A} = \overline{A}$$

3. Very imp Boolean theorems

- $A \cdot (B + C) = A \cdot B + A \cdot C \longrightarrow$ AND is distributive over 'OR'

- $A + (B \cdot C) = (A + B) \cdot (A + C) \longrightarrow$ OR is distributive over AND.

- $A + \bar{A}B = (A + \bar{A}) \cdot (A + B) = (A + B)$

- $\bar{A} + AB = (\bar{A} + A) \cdot (\bar{A} + B) = (\bar{A} + B)$

- $B + \bar{A}\bar{B} = (B + \bar{A}) \cdot (B + \bar{B}) = (\bar{A} + B)$

$$\begin{aligned}\bar{A}B + A &= (\bar{A} + A) \cdot (B + A) \\ &= (B + A) = (A + B)\end{aligned}$$

- $(A+B)(A+C) = A + B \cdot C$

$$(A+B)(A+C)(A+D) = A + B \cdot C \cdot D$$

- $(A+B+\bar{C})(A+\bar{B}+C)(A+B+C) = ?$

$$(A+B+\bar{C}) \left[(A+C) + \bar{B} \cdot B \right]$$

$$(A+B+\bar{C})(A+C)$$

$$= A + (B+\bar{C}) \cdot C$$

$$= (A+BC) = (A+B)(A+C)$$

$$= A + (B+\bar{C}) \cdot (\bar{B}+C)(B+C)$$

$$= A + (B+\bar{C})[C + \bar{B} \cdot B]$$

$$= A + (B+\bar{C}) \cdot C$$

$$= A + BC$$

$$\bullet (A + \overline{B} + \overline{C}) (\overline{A} + B + \overline{C}) (\overline{A} + \overline{B} + \overline{C}) =$$

$$(A + \overline{B} + \overline{C}) [(\overline{A} + \overline{C}) + B \cdot \overline{B}]$$

$$(A + \overline{B} + \overline{C}) (\overline{A} + \overline{C})$$

$$= \overline{C} + (A + \overline{B}) \cdot \overline{A}$$

$$= \overline{C} + \overline{A} \cdot \overline{B}$$

H.W.

$$Q.1. (A + \overline{B} + C) (A + \overline{B} + \overline{C}) (\overline{A} + B + \overline{C}) = ?$$

$$Q.2. (A + \overline{B} + C) (A + B \cdot D + C) (A + B + \overline{C}) = ?$$

$$Q.3. (A + BC) (A + B + \overline{C}) (A + \overline{B} + \overline{C}) = ?$$



2 Minute Summary

basics & Theorems.

Thank you

Chandan gupta sir pw.

GW
Soldiers !

